

Last Time:

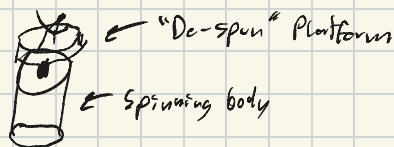
- Magnetic Torque
- B-dot / B-cross detumble

Today:

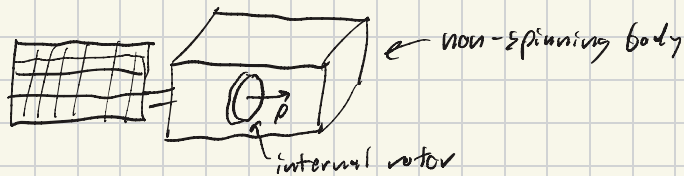
- Dual Spin / Momentum Bias
- Flat-Spin Recovery
- Actuator Jacobians

Dual-Spin / Momentum Bias:

- Dual Spin:



- Momentum Bias:



- Can achieve good ($\sim 1^\circ$) inertial pointing, but can't slew quickly
- Mathematically these are identical. They're both gyrostats:

$${}^B J^B \dot{\omega} + {}^B \dot{p} + \omega \times ({}^B J^B \omega + {}^B p) = \tau$$

Dual-Spin Stability:

- Assume rotor axis along $\hat{b}_z$
- Assume ${}^B J$ is diagonal:

$${}^B J = \begin{bmatrix} J_x & & \\ & J_y & \\ & & J_z \end{bmatrix}$$

- Assume a nominal spin ${}^B \omega = \begin{bmatrix} 0 \\ 0 \\ \omega_0 \end{bmatrix}$

$$\begin{cases} J_x \dot{\omega}_x + (J_z - J_y) \omega_0 \omega_y + p \omega_y = 0 \\ J_y \dot{\omega}_y + (J_x - J_z) \omega_0 \omega_x - p \omega_x = 0 \\ J_z \dot{\omega}_z + \dot{p} = 0 \end{cases}$$

- Spin axis dynamics are decoupled and assumed stabilized by a motor controller,
- Transverse Dynamics:

$$\frac{d}{dt} \begin{bmatrix} \omega_x \\ \omega_y \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & \frac{1}{J_x} [(J_y - J_z) \omega_0 - p] \\ \frac{1}{J_y} [(J_z - J_x) \omega_0 + p] & 0 \end{bmatrix}}_{\begin{bmatrix} 0 & a \\ b & 0 \end{bmatrix}} \begin{bmatrix} \omega_x \\ \omega_y \end{bmatrix}$$

* $\det(A - \lambda I) = 0 \Rightarrow \lambda = \pm \sqrt{ab} \Rightarrow$ we want $ab < 0$ for marginal stability.

$$\Rightarrow \begin{array}{cc} \rho > (J_x - J_s) \omega_0 & \text{and} \quad \rho > (J_y - J_s) \omega_0 \\ \rho < (J_x - J_s) \omega_0 & \text{OR and} \quad \rho < (J_y - J_s) \omega_0 \end{array}$$

Flat-Spin Recovery:

- Spacecraft can end up in an undesired major-axis spin (flat spin) due to malfunctions.
- We need a control law to recover the spacecraft's nominal minor-axis spin (especially for dual-spin spacecraft)
- Basic Idea: Minor-axis spin is a maximum-energy state for the system, so just pump kinetic energy into the system with the controller.

$$T = \frac{1}{2} \omega^T J \omega \Rightarrow \dot{T} = \omega^T J \dot{\omega}$$

$$J \dot{\omega} + \dot{\rho} + \omega \times (J \omega + \rho) = \tau$$

$$\Rightarrow \dot{\omega} = -J^{-1} (\dot{\rho} + \omega \times (J \omega + \rho) - \tau)$$

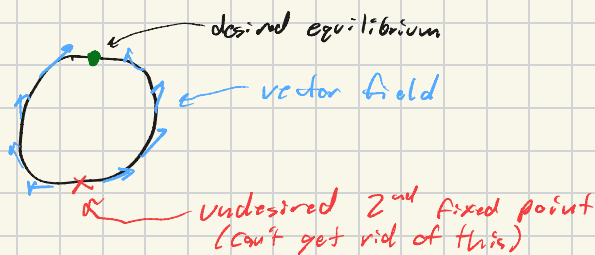
$$\begin{aligned} \Rightarrow \dot{T} &= -\omega^T \cancel{J} \overset{I}{J^{-1}} (\dot{\rho} + \omega \times \cancel{J} \overset{0}{J \omega + \rho}) - \tau \\ &= -\omega^T \dot{\rho} + \omega^T \tau \end{aligned}$$

* We want $\dot{T} \geq 0$, so a reasonable control law is:

$$\dot{\rho} = -K \omega \quad \text{or} \quad \tau = K \omega$$

↖ scalar gain ↗

- Can implement with rotors/wheels or thrusters/thrustors.
- Same control law also works as a nutation damper. Often implementing with a wheel offers better performance for less weight than a hardware fluid damper.
- This controller cannot guarantee the final sign of $\dot{\theta}$ since there are two minor-axis equilibria with the same energy.
- Can fix this with a more complicated Lyapunov function.
- Always have another fixed point at the "south pole":



- This is related to the "hairy ball theorem"

Actuator Jacobians:

- So far we have written our dynamics in terms of $\dot{p}$ and τ :

$$J\dot{\omega} + \dot{p} + \omega \times (J\omega + p) = \tau$$

- We need mappings between actuator commands (reaction wheel torques, thruster forces, etc.) and $\dot{p}$ and τ

- For thrusters:

$${}^B\chi_T = \sum_{i=1}^N {}^B r_i \times {}^B F_i = \sum_{i=1}^N ({}^B r_i \times {}^B a_i) u_i$$

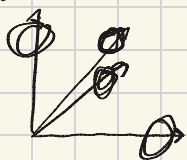
↑ Thruster axis unit vector
← Thruster force command
↑ Thruster position relative to CoM

$$\Rightarrow {}^B\chi_T = \underbrace{\begin{bmatrix} {}^B r_1 \times {}^B a_1 & {}^B r_2 \times {}^B a_2 & \dots & {}^B r_n \times {}^B a_n \end{bmatrix}}_{B_T} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$$

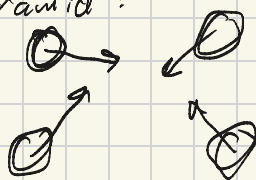
- Reaction Wheels:

- While 3 reaction wheels are needed for full control authority, most spacecraft have 4 or more for redundancy.
- Four wheels can be arranged in various ways to provide higher performance + single-fault tolerance.
- Some common choices:

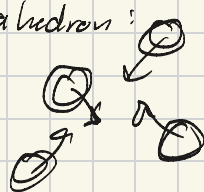
JPL:



Pyramid:



Tetrahedron:



- Wheel Jacobian

$${}^B p = \underbrace{\begin{bmatrix} {}^B a_1 & {}^B a_2 & \dots & {}^B a_n \end{bmatrix}}_{B_w} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \quad \begin{cases} \chi_w = B_w u_w \\ \dot{w} = u_w \end{cases}$$

↑ wheel spin-axis unit vectors
↑ individual wheel momenta

- To produce a desired torque:

$$\begin{bmatrix} \tau \end{bmatrix} = \begin{bmatrix} B \end{bmatrix} \begin{bmatrix} u \end{bmatrix}$$

"underdetermined"
linear system

* there are an infinite number of u vectors that can generate the same τ

$$\left. \begin{array}{l} \min_u \frac{1}{2} u^T u \\ \text{s.t. } \tau = Bu \end{array} \right\} \quad L = \frac{1}{2} u^T u + \lambda^T (Bu - \tau)$$

$$\Rightarrow \begin{array}{l} \nabla_u L = u + B^T \lambda \\ \nabla_\lambda L = Bu - \tau \end{array} \Rightarrow u = -B^T \lambda$$

$$\Rightarrow -BB^T \lambda - \tau = 0 \Rightarrow \lambda = -(BB^T)^{-1} \tau$$

$$\Rightarrow u - B^T (BB^T)^{-1} \tau = 0$$

$$\Rightarrow \boxed{u = B^T (BB^T)^{-1} \tau}$$

"Pseudo inverse" of B

* Pseudo inverse gives minimum 2-norm solution (usually corresponds to minimum overall power/energy). Other choices (e.g. 1-norm) are possible.

* Note: this is a different pseudoinverse than the one we saw previously for the "overdetermined" / "least-squares" case